

Returns to Representation: Girls' Youth Sports Participation Following the Founding of the WNBA

Tate Mason

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Abstract

This paper investigates the impact of a women's professional sports league on girls' youth sports participation. Using the Women's National Basketball Association's (WNBA) inaugural season in 1997 as treatment, I apply the methods of [CSA] methods to determine if there are any effects of the league's formation on the participation of girls in high school sports. Across four sports, I find modest increases in basketball and track participation, while cross-country and soccer do not exhibit any statistically significant effects.

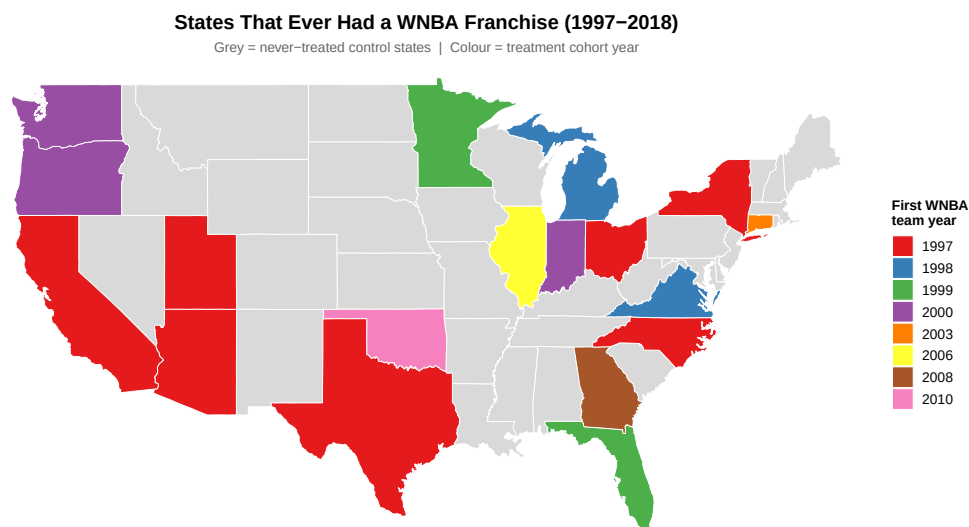
1 Introduction

In 1997, the Women’s National Basketball Association (WNBA) held its inaugural season, marking a significant milestone in the history of women’s professional athletics. The league’s formation provided a new platform through which female athletes could showcase their talents and serve as role models as professionals rather than amateurs. The WNBA’s establishment was not only a significant moment for professional sports, but held plausible implications for sports participation for the next generation of female athletes. This paper investigates the impact of the WNBA’s formation and evolution on girls’ youth sports participation, particularly in high school athletics.

Professional athletes can serve as role models for the youth, and the presence of a professional sports league can influence youth participation in several ways. First, the visibility of professional athletes can inspire engagement in sports at a young age. Second, the presence of a professional team can increase the availability of resources and infrastructure (i.e. facilities, coaching, etc.) for youth sports in the local community. Third, the presence of a professional team can increase the social acceptance and cultural relevance of a given sport, which can encourage participation.

The question of whether the presence of a professional sports league can influence youth participation is of interest to both the league as well as educators. For the league, understanding how the presence of a professional team in a given market can influence youth participation can inform marketing, outreach, and even future expansion decisions. For educators, understanding how to encourage youth sports participation via role models can promote physical activity and the associated benefits for health and well-being. Building upon the work of [Reilly’rod], I apply two-way fixed effects methods to data on high school sports participation from the National Federation of State High School Associations (NFHS) to ascertain whether the presence of a WNBA team in a given state is associated with increases in girls’ sports participation. I find modest increases in basketball and track participation, while cross-country and soccer do not exhibit any statistically significant effects. I then apply a synthetic differences-in-differences method to further investigate the effects of a state having a WNBA team on high school sports participation, and find more varied results.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature on the effects of professional sports leagues on youth sports participation. Section 3 describes the data and estimation strategy. Section 4 presents the results, while Section 5 discusses findings. Finally, Section 6 concludes.



2 Literature

3 Methodology

3.1 Data

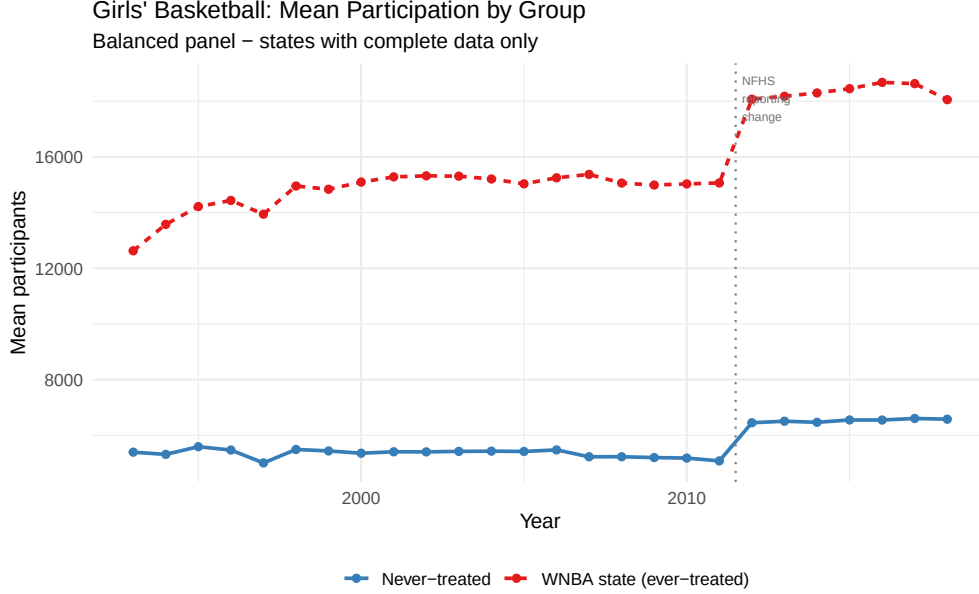
The primary data source for this analysis is National Federation of State High School Associations (NFHS) data on high school sports participation. The NFHS collects data on the number of participants in various sports at the high school level across the United States. I focus on four sports: basketball, track and field, cross-country, and soccer. The treatment variable is an indicator for whether a state has a WNBA team in a given year, which I construct using data on WNBA team locations and years of operation. Data on WNBA team locations and years of operation are obtained from the official WNBA website. Figure 3.1 shows the expansion path of the WNBA from its inaugural season in 1997 to 2010, the last year of expansion considered in this analysis. From 2010 to present, the WNBA has expanded to a total of 15 teams, with a further three to be added by 2030.

Data from NFHS spans 1993 to 2016, allowing for a pre-treatment period of four years and a post-treatment period of 6 years. The data is aggregated at the state-year level, and I balance the panel so that each state has observations for all years in the sample. The final dataset includes 47 states and 24 years, results in a total of 1,128 state-year observations. ?? provides summary statistics for the key variables in the analysis.

TABLE 1 HERE

?? provides summary statistics for treated states.

TABLE 2 HERE



3.2 Estimation

Following [CSA], I estimate the following differences-in-differences model:

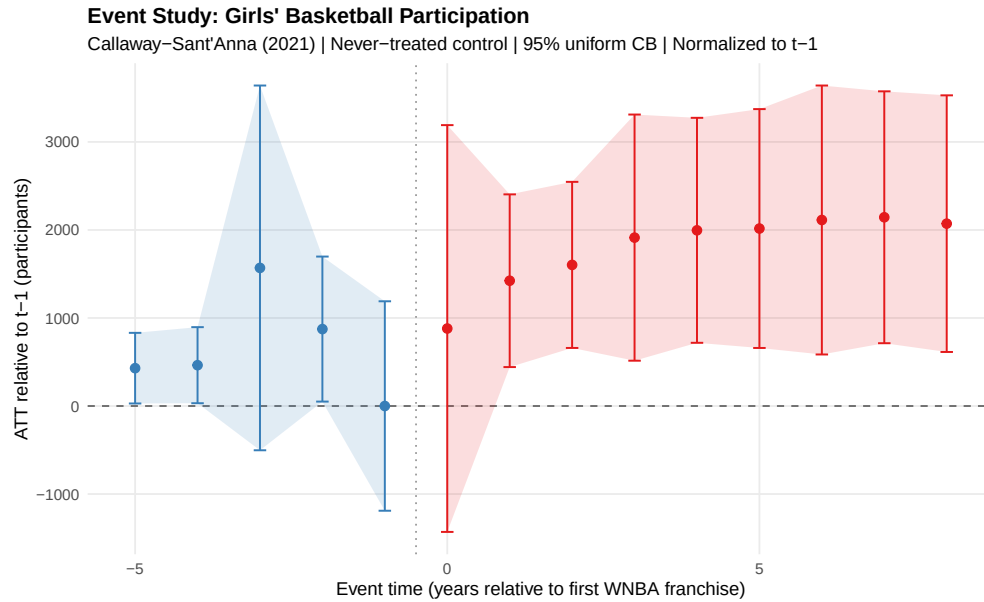
$$\hat{ATT}_{gt} = (Y_t - Y_{g-1}|G = g) - (Y_t - Y_{g-1}|WNBA = 0) \quad (1)$$

where Y_t is the participation in a given sport in year t for a state which receives a WNBA team in treatment cohort g , and Y_{g-1} is the participation in the year prior to treatment for that cohort. The second term is the same difference but for states that do not ever receive a WNBA team. I then aggregate these estimates across cohorts to obtain an overall estimate of the effect of having a WNBA team. Standard errors are clustered at the state level to account for serial correlation within states over time.

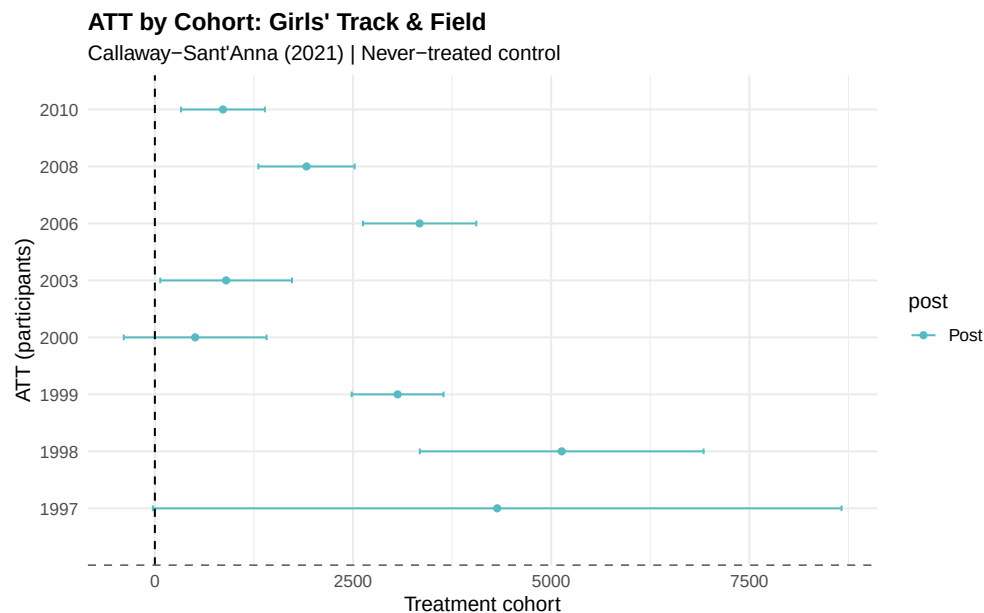
I also apply the synthetic differences-in-differences method of [arkhangelsky] to test for robustness of the results. This method creates a synthetic control group for each treated cohort by weighting the control states to match the pre-treatment trends of the treated states. This allows for a more flexible comparison between treated and control states, helping to address potential violations of the parallel trends assumption needed for the staggered differences-in-differences approach.

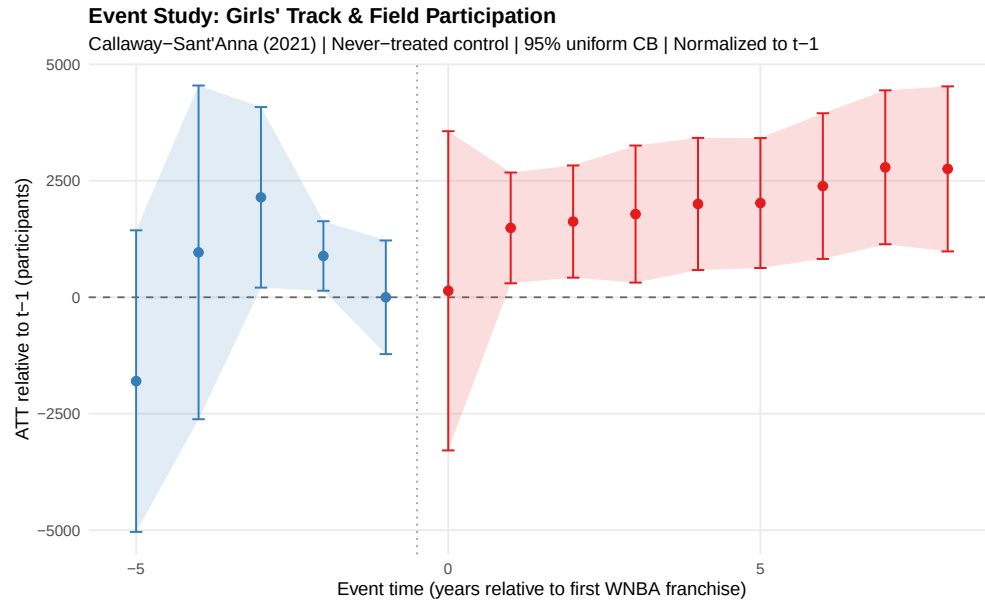
4 Results

From Figure 4, we can see that the pre-treatment coefficients are relatively flat and close to zero, suggesting that there are no significant differences in basketball participation trends between treated and control states prior to the treatment. This provides some evidence in support of the parallel trends assumption for basketball participation. To be transparent, there are a couple of years with wide confidence bands, lending to some uncertainty about the validity of the result, but given the



overall picture, I am comfortable proceeding under the assumption of parallel trends for basketball participation. When looking at the other sports, soccer has no discernible pre-treatment nor does cross-country, but track and field appears on a mostly upward trend until two years before treatment. This leads to some dubiousness about the validity of parallel trends for track and field. Again, however, the standard errors are wide for track in the early event years, so I do not think there is much evidence to suggest a conclusion either way.





5 Robustness

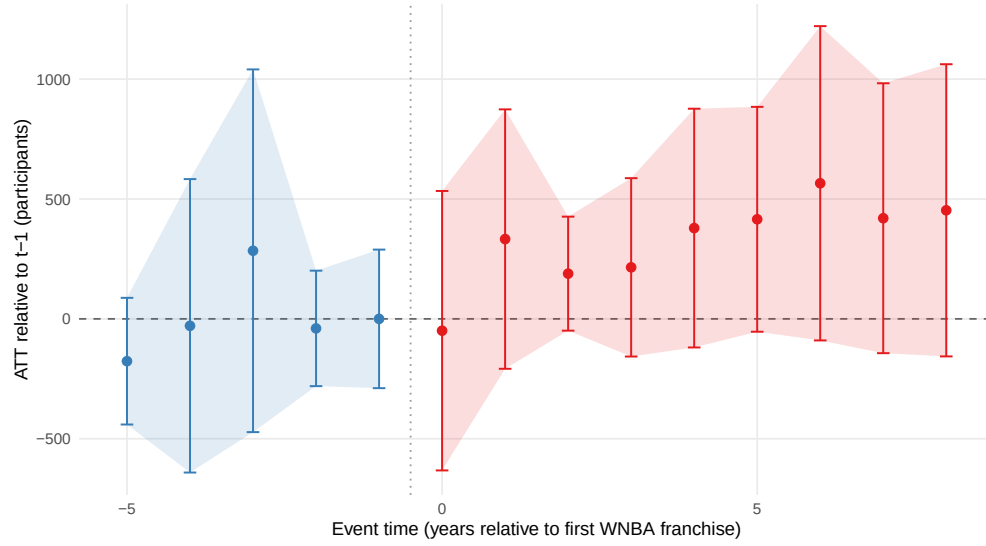
6 Discussion

7 Conclusion

Tables and Figures

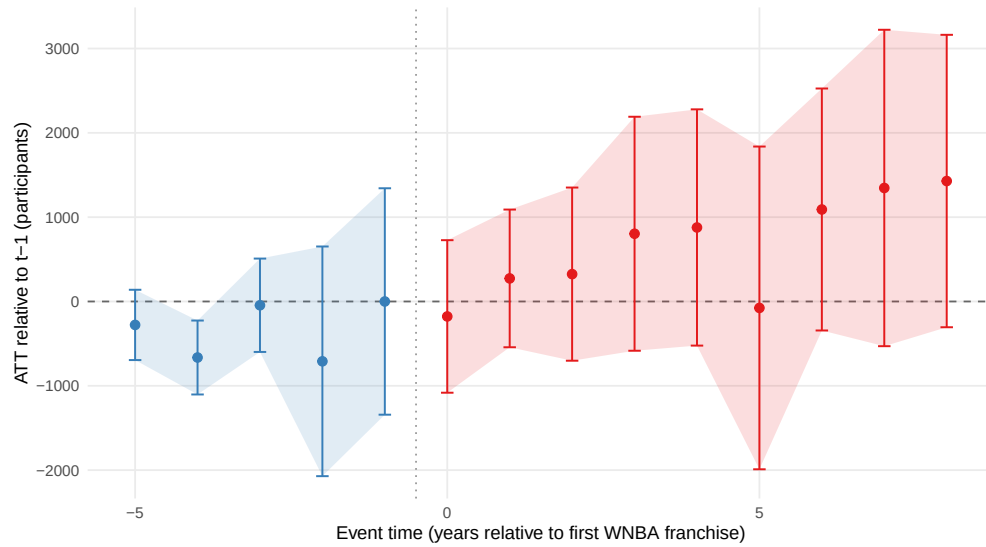
Event Study: Girls' Cross Country Participation

Callaway-Sant'Anna (2021) | Never-treated control | 95% uniform CB | Normalized to t-1



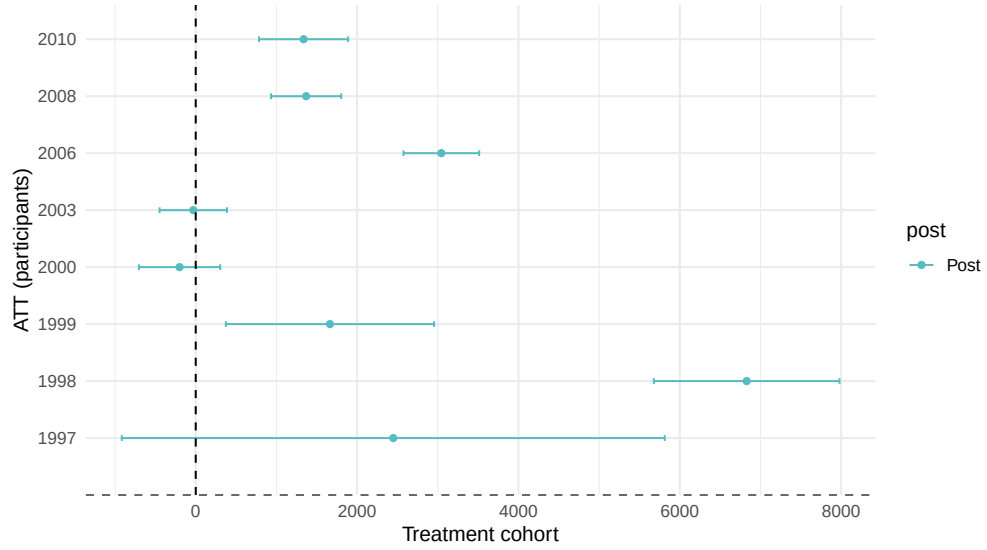
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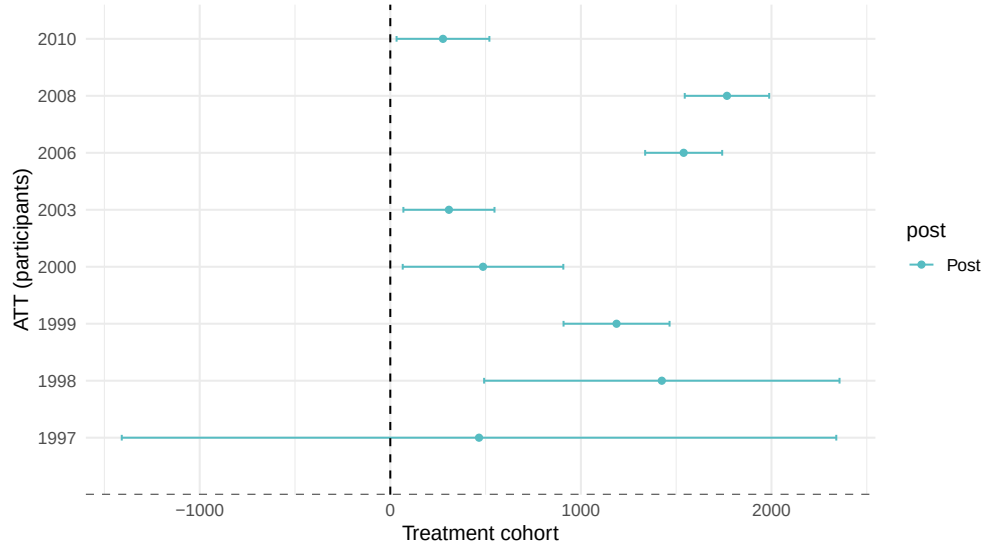
ATT by Cohort: Girls' Basketball

Callaway-Sant'Anna (2021) | Never-treated control



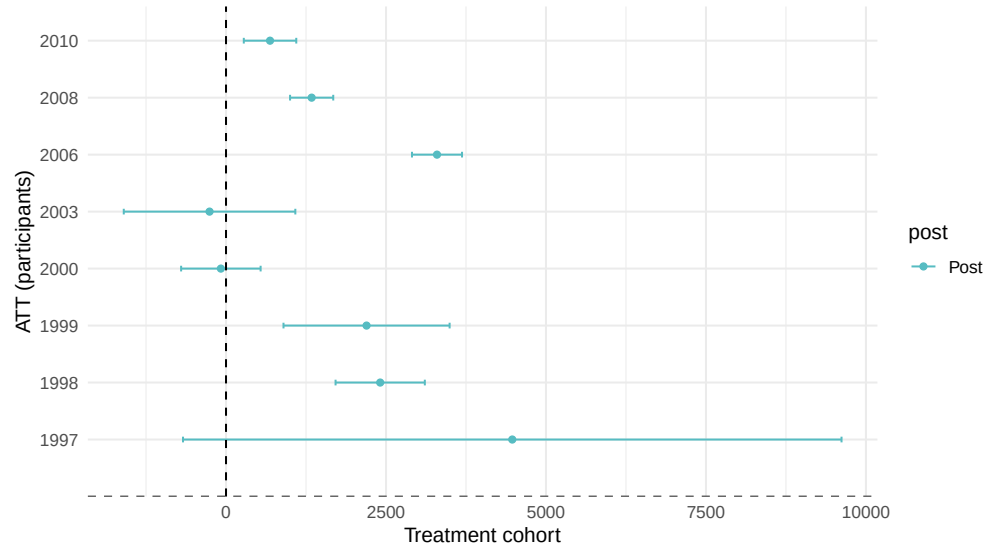
ATT by Cohort: Girls' Cross Country

Callaway-Sant'Anna (2021) | Never-treated control



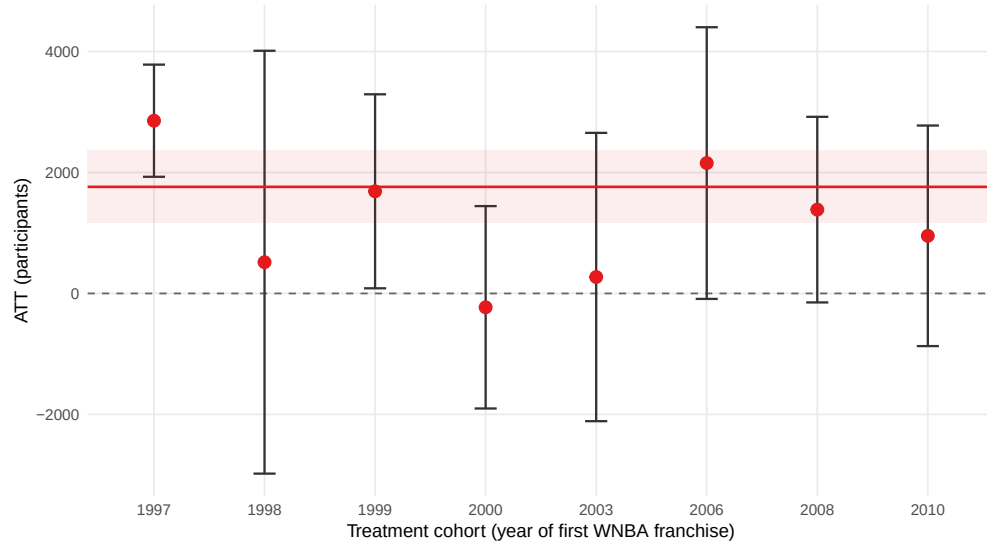
ATT by Cohort: Girls' Soccer

Callaway-Sant'Anna (2021) | Never-treated control



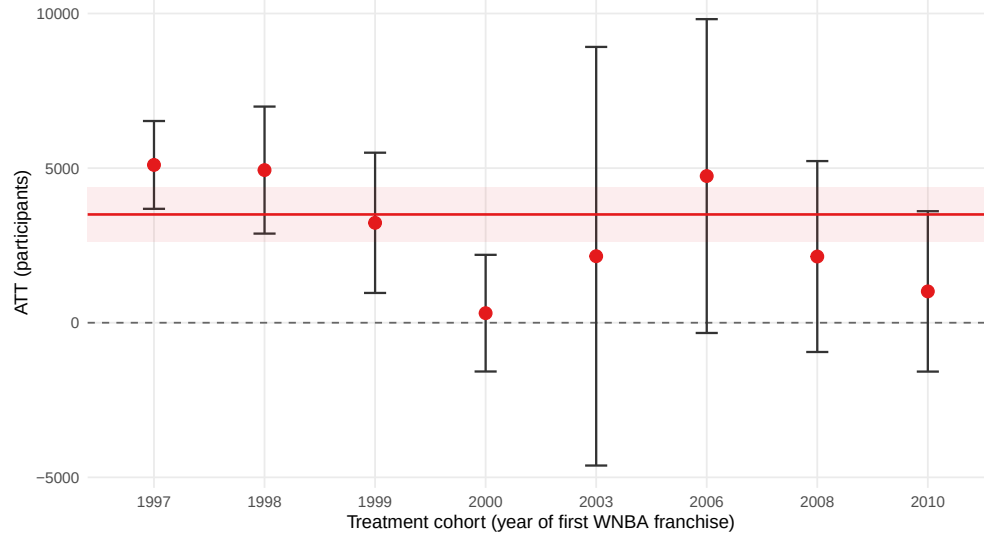
Synthetic DiD by Cohort: Girls' Basketball

Arkhangelsky et al. (2021) | Red band = pooled 95% CI | Red line = weighted average ATT



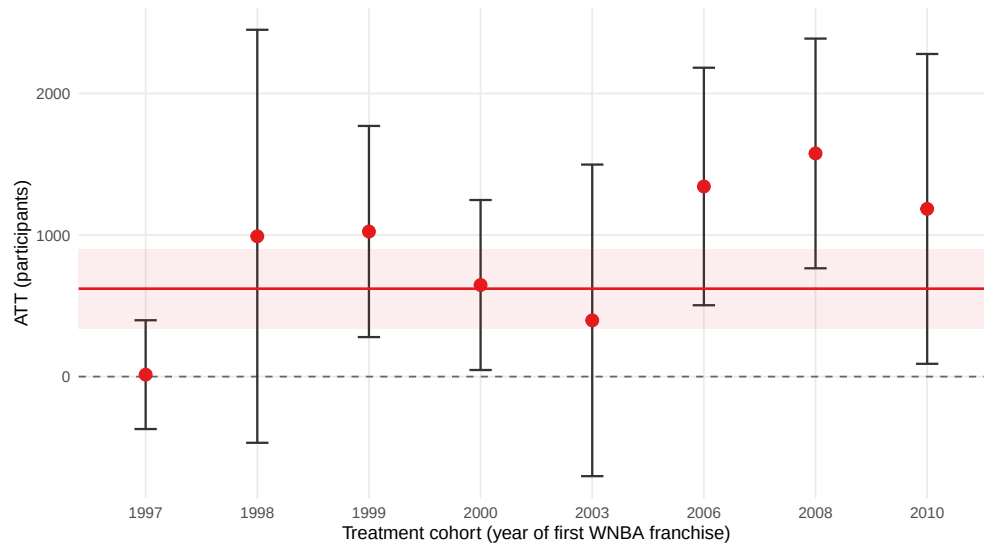
Synthetic DiD by Cohort: Girls' Track & Field

Arkhangelsky et al. (2021) | Red band = pooled 95% CI | Red line = weighted average ATT



Synthetic DiD by Cohort: Girls' Cross Country

Arkhangelsky et al. (2021) | Red band = pooled 95% CI | Red line = weighted average ATT



Synthetic DiD by Cohort: Girls' Soccer

Arkhangelsky et al. (2021) | Red band = pooled 95% CI | Red line = weighted average ATT

